

Analysis of the Research Proposal

Quick Summary

The proposal aims to build a **Hybrid AI + Explainable AI (XAI)** framework that links **soil microbiome data** to **mango yield/health** in Malihabad (U.P.). The pipeline is: *sampling* → *metagenomic sequencing* → *data integration* → *ML/DL modeling* → *SHAP/LIME explainability* → *decision support system* → *field validation*.

Strengths

- Clear multidisciplinary workflow (microbiology + data science + agronomy)
- Strong location specificity (Malihabad = famous Dashehari mango belt, alluvial soil)
- Good use of XAI for actionable, interpretable outputs

Gaps you should address before dataset creation

Issue	Recommendation
Sample size not defined	Specify n (e.g., 150–300 samples) for ML statistical power
Sequencing target unclear	Define 16S rRNA (bacteria/archaea) + ITS (fungi) regions
No baseline mango dataset	A synthetic/dummy dataset is needed for pipeline prototyping  (your request)
Minor errors	SDG 3 = "Good Health & Well-being" (Climate Action = SDG 13); budget lacks totals/justification
Target variable undefined	Clarify if predicting <i>yield (regression)</i> or <i>disease risk/nutrient class (classification)</i>

Microbiome–Mango Dataset Parameters

Based on the methodology (Sections 3.1.1–3.1.5), here are the parameters organized into **5 logical blocks**, with realistic units and value ranges for Malihabad alluvial soils.

Block 1 — Sample & Site Metadata

Parameter	Type	Range / Values	Unit
Sample_ID	ID	S0001–S0300	—
Orchard_ID	Categorical	ORCH_01–ORCH_30	—
Village	Categorical	Malihabad, Mall, Kakori, Rahimabad	—
Latitude	Float	26.85–27.05	°N
Longitude	Float	80.65–80.90	°E
Mango_Variety	Categorical	Dashehari, Langra, Chausa, Safeda	—
Tree_Age	Integer	5–60	years
Soil_Depth	Categorical	0–15, 15–30	cm
Sampling_Season	Categorical	Pre-monsoon, Post-monsoon, Winter	—
Management	Categorical	Organic, Conventional, Integrated	—

Block 2 — Soil Physicochemical Parameters

Parameter	Range (Malihabad)	Unit
pH	6.5 – 8.5	—
EC (Electrical Conductivity)	0.2 – 0.9	dS/m
Organic_Carbon	0.3 – 0.9	%
Total_Nitrogen	150 – 400	kg/ha
Available_P	8 – 30	kg/ha
Available_K	120 – 350	kg/ha
Soil_Moisture	8 – 35	%
Soil_Temperature	18 – 38	°C

Bulk_Density	1.2 – 1.6	g/cm ³
CEC	8 – 25	cmol/kg
Sand / Silt / Clay	sums to 100	%
Zn / Fe / Mn / Cu	0.4–2.0 / 3–20 / 1–10 / 0.2–3	ppm
C_N_Ratio	8 – 15	ratio

Block 3 — Climate / Weather Variables

Parameter	Range	Unit
Air_Temp_Avg	15 – 42	°C
Rainfall	0 – 300	mm/month
Humidity	30 – 90	%
Solar_Radiation	12 – 26	MJ/m ² /day

Block 4 — Microbiome Data (Core of the Study)

(a) Diversity indices (derived from OTU table)

Parameter	Range	Notes
OTU_Count	800 – 3000	total OTUs
Shannon_Index	2.5 – 6.0	α-diversity
Simpson_Index	0.6 – 0.99	dominance
Chao1_Richness	900 – 3500	estimated richness

Pielou_Evenness 0.5 – 0.95 evenness

(b) Relative abundance of taxa (%) — sum ≈ 100 within each domain

Bacterial Phyla	Fungal Phyla	Archaea
Proteobacteria (20–40)	Ascomycota (40–70)	Thaumarchaeota (1–8)
Actinobacteria (15–30)	Basidiomycota (10–30)	Euryarchaeota (0.5–5)
Acidobacteria (8–20)	Glomeromycota/AMF (2–15)	
Firmicutes (5–15)	Mortierellomycota (1–10)	
Bacteroidetes (5–15)	Zygomycota (0.5–5)	

(c) Functional / beneficial & pathogenic groups (key predictors)

Parameter	Range	Role
Nitrogen_Fixers (Azotobacter, Rhizobium)	1×10 ⁴ – 1×10 ⁷	CFU/g — beneficial
Phosphate_Solubilizers_PSB	1×10 ⁴ – 1×10 ⁶	CFU/g — beneficial
Potassium_Solubilizers	1×10 ³ – 1×10 ⁵	CFU/g — beneficial
Mycorrhizae_AMF	50 – 400	spores/100g — beneficial
Trichoderma (biocontrol)	1×10 ³ – 1×10 ⁵	CFU/g — beneficial
Pseudomonas_PGPR	1×10 ⁴ – 1×10 ⁶	CFU/g — beneficial
Fusarium (pathogen)	0 – 1×10 ⁵	CFU/g — harmful
Pathogen_Load_Index	0 – 1	normalized risk

Block 5 — Engineered Features & Targets

Parameter	Description	Type
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Microbial_Richness_Score	composite of diversity indices	Feature
Nutrient_Balance_Ratio	f(N, P, K)	Feature
Soil_Health_Index	composite 0–100	Feature
Mango_Yield	8 – 25 (target)	kg/tree → Regression
Disease_Risk	Low / Medium / High	Classification
Nutrient_Availability	Deficient / Optimal / High	Classification
